

AMAZON SELLER SQL DATA ANALYTICS PROJECT

End-to-End SQL Analysis of 1,000 E-commerce Orders

Tools: SQL (SQLite) • Excel/CSV Dataset • Data Analysis • Business Insights

Dataset: amazon_seller.csv | **Records:** 1,000 | **Period:** Jan–Dec 2024

Prepared as a portfolio-ready Data Analyst SQL project

1. Project Overview

This project analyzes an e-commerce seller dataset containing 1,000 orders across products, categories, cities, seller states, payment methods, discounts and order statuses. The goal is to demonstrate practical SQL skills used by a Data Analyst: filtering, sorting, aggregation, GROUP BY, HAVING-style logic, ranking with ORDER BY/LIMIT, and translating query results into business recommendations.

Business Questions

- Which categories and products generate the most revenue?
- Which cities are the strongest customer markets?
- Which payment methods are used most often?
- What orders have the highest discounts and order values?
- Which orders have quantity greater than 2?
- What is the recent order activity from Mumbai?
- What is the overall distribution of payment methods and order statuses?

Dataset Structure

Column	Description
order_id	Unique order identifier
order_time	Order date and time
timeframe	Hourly order window
product_name	Product purchased
category	Product category
seller_id / seller_state	Seller identifier and state
customer_area / customer_city	Customer location
quantity	Units purchased
unit_price	Original unit price
discount_percent	Discount applied
order_value	Final order value
payment_method	Payment channel
order_status	Delivered / Returned / Cancelled / Pending
delivery_date	Delivery date where available

2. Executive KPI Snapshot

KPI	Result	Interpretation
Total Orders	1,000	Dataset contains 1,000 transactions
Total Revenue	■1,794,152	Total order value across all records
Units Sold	1,357	Total quantity purchased
Average Order Value	■1,794	Average revenue per order
Average Discount	23.9%	Average discount across orders
Delivered Rate	74.9%	Share of orders marked Delivered
Returned Rate	9.4%	Share of orders marked Returned
Cancelled Orders	99	Orders marked Cancelled
Pending Orders	58	Orders still marked Pending

Key Findings

Electronics is the largest revenue category at ■823,459, representing 45.9% of total revenue. **Mumbai** is the highest-revenue customer city at ■287,461. **UPI** is the most-used payment method with 308 orders. The dataset shows a 9.4% return rate, which is an important operational metric to investigate.

3. Core SQL Analysis Queries

3.1 Inspect the dataset

```
SELECT *  
FROM amazon_seller;
```

3.2 Top 10 highest-discount orders

```
SELECT discount_percent, category, order_value, product_name  
FROM amazon_seller  
ORDER BY discount_percent DESC  
LIMIT 10;
```

3.3 Top 5 highest-value orders

```
SELECT order_value, product_name, category  
FROM amazon_seller  
ORDER BY order_value DESC  
LIMIT 5;
```

3.4 Orders where quantity is greater than 2

```
SELECT quantity, customer_city, product_name  
FROM amazon_seller  
WHERE quantity > 2;
```

3.5 Top 5 orders where quantity is greater than 2

```
SELECT quantity, customer_city, product_name  
FROM amazon_seller  
WHERE quantity > 2  
ORDER BY quantity DESC  
LIMIT 5;
```

4. Location & Payment Analysis

4.1 All Mumbai orders

```
SELECT *  
FROM amazon_seller  
WHERE customer_city = 'Mumbai';
```

4.2 Five most recent Mumbai orders

```
SELECT *  
FROM amazon_seller  
WHERE customer_city = 'Mumbai'  
ORDER BY order_time DESC  
LIMIT 5;
```

4.3 Cash on Delivery orders

```
SELECT *  
FROM amazon_seller  
WHERE payment_method = 'Cash on Delivery';
```

4.4 UPI orders

```
SELECT *  
FROM amazon_seller  
WHERE payment_method = 'UPI';
```

4.5 Total payment records

```
SELECT COUNT(payment_method) AS total_payment_records  
FROM amazon_seller;
```

4.6 Payment method distribution

```
SELECT payment_method,  
       COUNT(payment_method) AS payment_count  
FROM amazon_seller  
GROUP BY payment_method  
ORDER BY payment_count DESC;
```

5. Query Results & Validation

Payment Method Counts

Payment Method	Orders	Revenue
UPI	308	■598,488
Credit Card	191	■331,442
Debit Card	186	■345,447
Cash on Delivery	146	■251,921
Net Banking	99	■151,440
Amazon Pay	70	■115,414

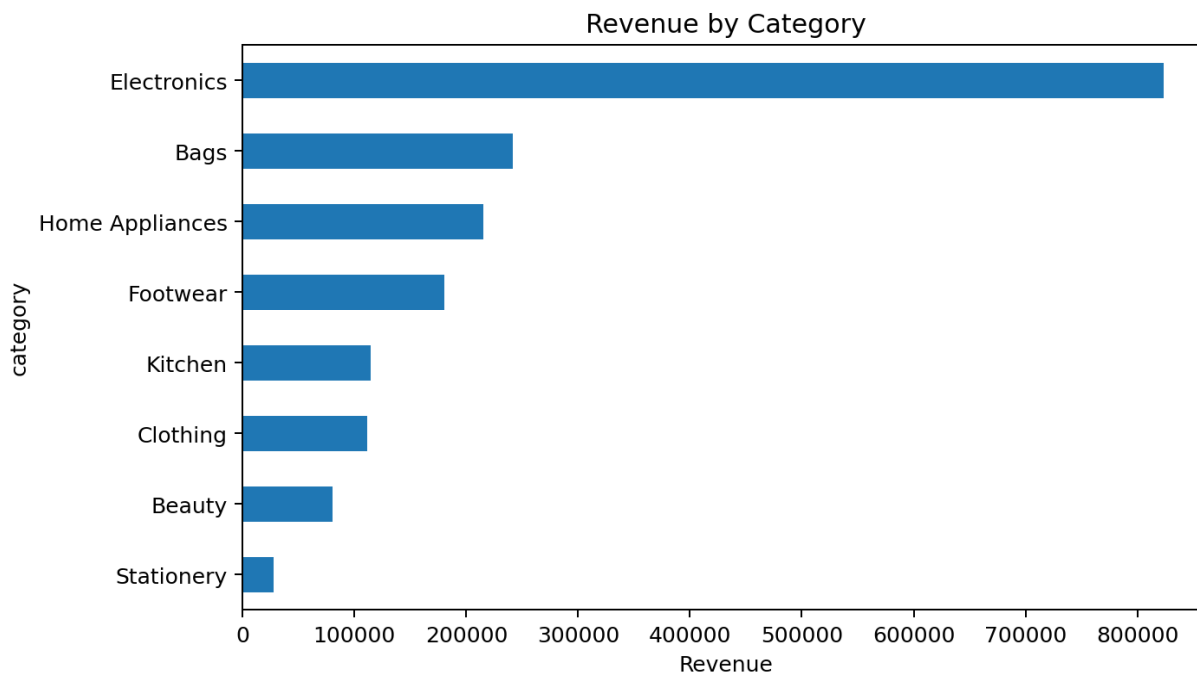
Top 5 Highest-Value Orders

Order Value	Product	Category
■13,524	Realme Smart Watch	Electronics
■12,084	Samsung Galaxy Buds	Electronics
■11,283	boAt Rockerz 255	Electronics
■10,245	Fire TV Stick	Electronics
■10,149	Samsung Galaxy Buds	Electronics

Top Quantity > 2 Orders

Quantity	Customer City	Product
3	Pune	Puma Running Shoes
3	Kolkata	Mamaearth Shampoo
3	Hyderabad	Realme Smart Watch
3	Kolkata	Philips Hair Dryer
3	Bengaluru	Lakme Face Cream
3	Kolkata	Mamaearth Shampoo
3	Delhi	Bata Formal Shoes
3	Pune	boAt Rockerz 255
3	Pune	Puma Running Shoes
3	Kolkata	Mamaearth Shampoo

6. Category & Product Performance

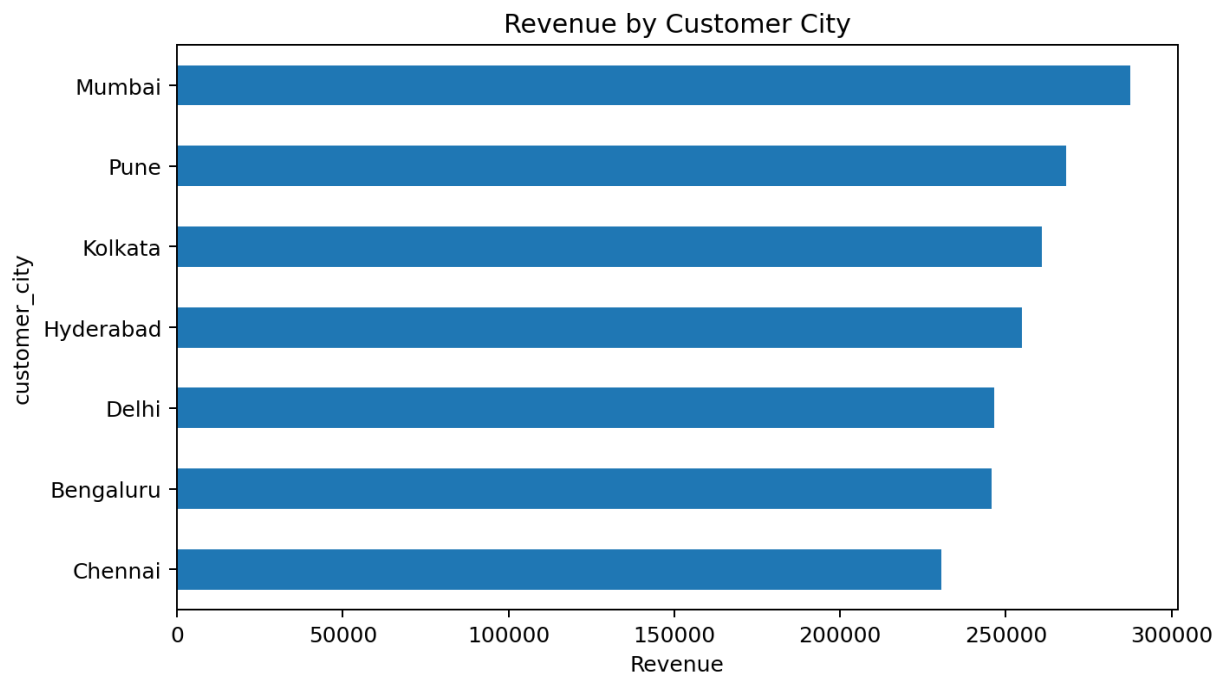


Category	Orders	Units	Revenue	Revenue Share
Electronics	297	412	■823,459	45.9%
Bags	103	145	■241,456	13.5%
Home Appliances	87	116	■215,457	12.0%
Footwear	96	130	■180,657	10.1%
Kitchen	100	128	■114,135	6.4%
Clothing	96	126	■111,441	6.2%
Beauty	126	174	■80,168	4.5%
Stationery	95	126	■27,379	1.5%

Analytical insight: Electronics leads with 297 orders and ■823,459 revenue. It is the most important category for inventory, availability and merchandising decisions in this dataset.

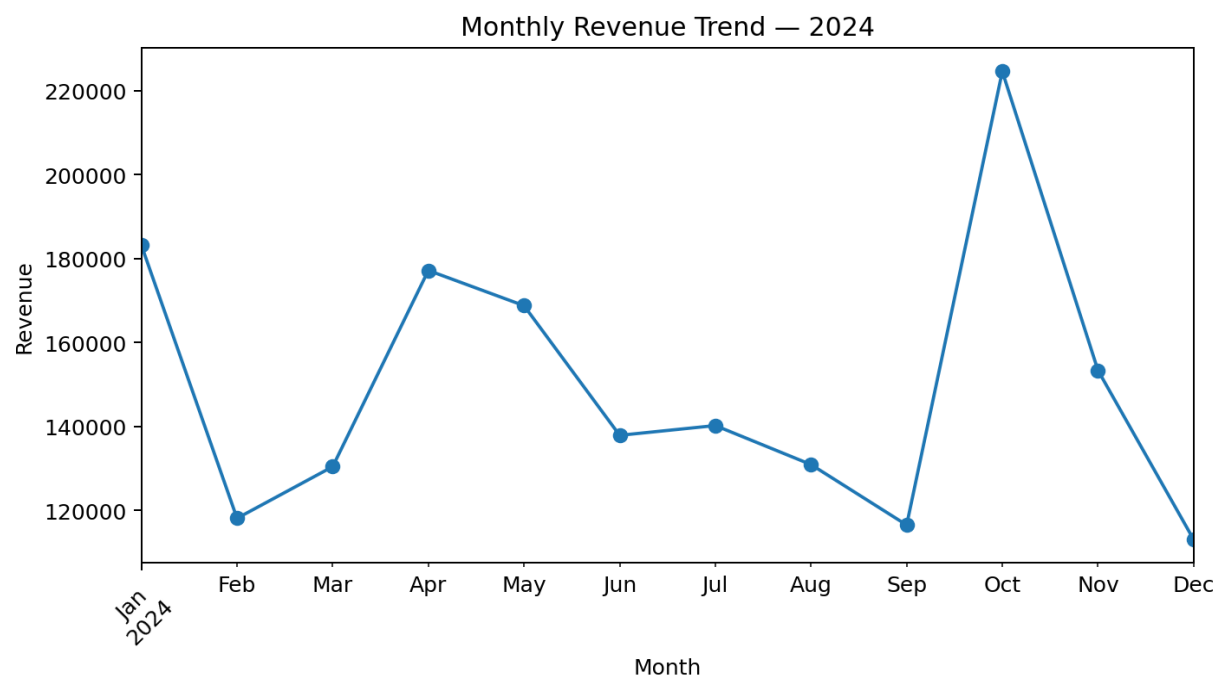
7. Geography & Time-Series Analysis

Revenue by Customer City

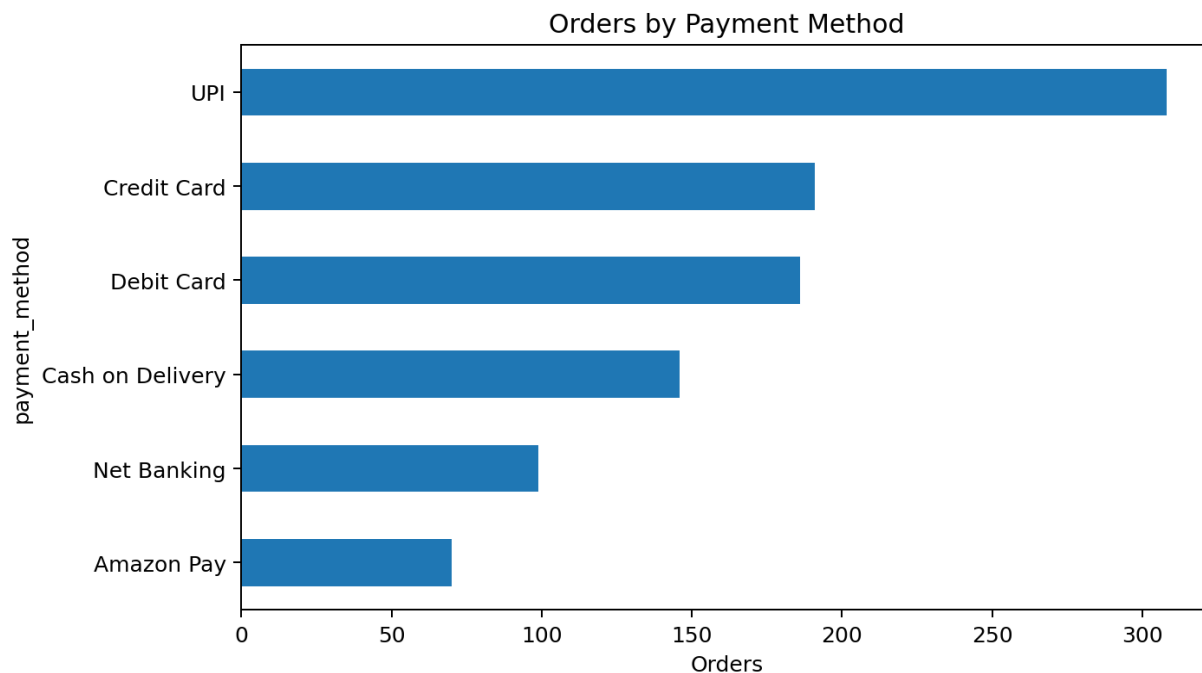


City	Orders	Revenue
Mumbai	154	■287,461
Pune	143	■268,313
Kolkata	136	■260,804
Hyderabad	149	■254,924
Delhi	138	■246,475
Bengaluru	141	■245,720
Chennai	139	■230,455

Monthly Revenue Trend



8. Payment & Order Status Insights



Order Status Distribution

Status	Orders	Revenue	Share of Orders
Delivered	749	■1,361,177	74.9%
Cancelled	99	■150,902	9.9%
Returned	94	■180,541	9.4%
Pending	58	■101,532	5.8%

Payment insight: UPI accounts for 30.8% of orders and is the strongest payment channel. This supports maintaining a frictionless UPI checkout experience. **Operations insight:** 193 orders are Returned or Cancelled, so reducing post-order failure can materially improve realized revenue and customer experience.

9. Business Insights & Recommendations

Important data-quality correction: The uploaded dataset is India-focused, with cities such as Mumbai, Pune, Kolkata, Hyderabad and Delhi. Therefore, recommendations should not refer to the US or New York unless a separate US dataset is added.

Recommendation 1 — Prioritize Electronics

Electronics generates ■823,459, or 45.9% of total revenue. Increase availability of high-performing electronics, monitor stock-outs, and prioritize these products in campaigns and cross-selling.

Recommendation 2 — Strengthen High-Revenue Cities

Mumbai leads customer revenue at ■287,461, followed by Pune at ■268,313. Focus local inventory, promotions and delivery capacity on the highest-value cities.

Recommendation 3 — Investigate Returns & Cancellations

The return rate is 9.4% and cancelled orders represent 9.9% of all orders. Segment these by product, category, seller and city to identify root causes such as product quality, inaccurate descriptions, pricing or delivery issues.

Recommendation 4 — Use Payment Behavior

UPI is the dominant payment method. Keep UPI prominent while maintaining alternative payment methods for customers who prefer cards, COD, net banking or Amazon Pay.

Recommendation 5 — Product-Level Investigation

Do not label products as underperforming from a single revenue ranking alone. Compare order count, units, discount, return rate and category benchmarks before changing pricing or inventory strategy.

10. Advanced SQL Queries for Portfolio Expansion

10.1 Revenue by category

```
SELECT category,
       COUNT(*) AS orders,
       SUM(quantity) AS units,
       SUM(order_value) AS revenue,
       ROUND(AVG(order_value), 2) AS avg_order_value
FROM amazon_seller
GROUP BY category
ORDER BY revenue DESC;
```

10.2 Revenue by customer city

```
SELECT customer_city,
       COUNT(*) AS orders,
       SUM(order_value) AS revenue
FROM amazon_seller
GROUP BY customer_city
ORDER BY revenue DESC;
```

10.3 Product performance

```
SELECT product_name,
       COUNT(*) AS orders,
       SUM(quantity) AS units,
       SUM(order_value) AS revenue
FROM amazon_seller
GROUP BY product_name
ORDER BY revenue DESC;
```

10.4 Order-status analysis

```
SELECT order_status,
       COUNT(*) AS orders,
       SUM(order_value) AS revenue,
       ROUND(COUNT(*) * 100.0 / (SELECT COUNT(*) FROM amazon_seller), 2) AS order_share_pct
FROM amazon_seller
GROUP BY order_status
ORDER BY orders DESC;
```

10.5 Return rate by category

```
SELECT category,
       COUNT(*) AS total_orders,
       SUM(CASE WHEN order_status = 'Returned' THEN 1 ELSE 0 END) AS returned_orders,
       ROUND(
         SUM(CASE WHEN order_status = 'Returned' THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2
       ) AS return_rate_pct
FROM amazon_seller
GROUP BY category
ORDER BY return_rate_pct DESC;
```

10.6 Top products by revenue with minimum order volume

```
SELECT product_name,
       COUNT(*) AS orders,
       SUM(order_value) AS revenue
FROM amazon_seller
GROUP BY product_name
HAVING COUNT(*) >= 10
ORDER BY revenue DESC
LIMIT 10;
```

11. Portfolio & Resume Presentation

Project Description

Amazon Seller SQL Data Analytics Project — Analyzed 1,000 e-commerce transactions using SQLite SQL to identify category, product, city, payment and order-status trends. Built business insights around revenue concentration, customer geography, payment behavior and operational performance.

Skills Demonstrated

- SQL: SELECT, WHERE, ORDER BY, LIMIT, COUNT, GROUP BY, CASE WHEN, HAVING and subqueries
- Data cleaning and validation using a CSV-based dataset
- KPI analysis: revenue, orders, units, average order value and return rate
- Business analysis: category, product, city, payment and order-status performance
- Data storytelling: converting query results into actionable recommendations

Resume-Ready Achievement Bullets

- Analyzed 1,000 e-commerce orders using SQL and identified Electronics as the leading category, contributing 45.9% of total revenue.
- Evaluated customer geography and found Mumbai to be the highest-revenue city at ₹287,461, supporting targeted inventory and marketing decisions.
- Analyzed payment behavior and found UPI to be the most-used channel with 308 orders, supporting checkout optimization recommendations.
- Assessed order outcomes and quantified a 9.4% return rate and 9.9% cancellation rate to highlight operational improvement opportunities.

Suggested GitHub Project Structure

```
amazon-seller-sql-analysis/  
├── data/  
│   └── amazon_seller.csv  
├── sql/  
│   └── amazon_seller_analysis.sql  
├── screenshots/  
│   └── query_results/  
│       └── database/  
├── README.md  
└── Amazon_Seller_SQL_Data_Analytics_Project.pdf
```

12. Evidence & Sample Results

The screenshots supplied with the project show the SQL editor, amazon_seller table, query history and result grids. The examples below are validated against the uploaded CSV so the portfolio document is based on the actual dataset rather than placeholder numbers.

Sample: Five most recent Mumbai orders

Order ID	Order Time	Product	Category	Customer Area
AMZ000789	2024-12-28 12:07	Pigeon Pressure Cooker	Kitchen	Thane
AMZ000221	2024-12-26 12:07	Fire TV Stick	Electronics	Andheri
AMZ000027	2024-12-25 11:19	American Tourister Trolley	Bags	Dadar
AMZ000483	2024-12-24 12:03	Milton Water Bottle	Kitchen	Thane
AMZ000270	2024-12-23 07:05	Mamaearth Shampoo	Beauty	Borivali

Sample: Highest-discount orders

Discount	Category	Order Value	Product
50%	Electronics	■1,380	Samsung Galaxy Buds
50%	Bags	■1,360	American Tourister Trolley
50%	Electronics	■1,624	Samsung Galaxy Buds
50%	Bags	■1,010	American Tourister Trolley
50%	Kitchen	■310	Pigeon Pressure Cooker
50%	Electronics	■6,930	Samsung Galaxy Buds
50%	Electronics	■2,210	Samsung Galaxy Buds
50%	Clothing	■644	Levi's T-Shirt

Final takeaway: The project demonstrates a complete SQL-to-insight workflow: inspect the data, ask business questions, write SQL, validate outputs, summarize trends and convert findings into practical business actions. This is suitable as a fresher Data Analyst portfolio project.